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Proposal of a Modified Hash Algorithm to Increase Blockchain Security

Alaa Abid Muslam Abid Ali[a,b], Manar Joundy Hazar[a,b], Mohamed Mabrouk[a,*], Mounir Zrigui[a]

[a] *University of Monastir, Research Laboratory in Algebra, Numbers Theory and Intelligent System,Tunisia*
[b] *Al-Qadisiyah University, Al-Qadisiyah, Iraq*

Abstract

Blockchain technology provides a novel method for storing information, performing tasks and functions, and establishing trust between participating nodes. Despite the fact that blockchain technology has received significant attention in a variety of application contexts over the last few years, the issue of privacy and security remains the primary focus of blockchain discussions. In Blockchain, the cryptographic hash function is used to secure message blocks and connect blocks in a chain. In summary, each block in the blockchain contains its own block hash as well as a hash of the block before it. This enables them to create a cryptographically secure linear chain of blocks.With blockchain technology, there is a new way to store information, carry out tasks and functions, and build trust between the nodes that are taking part. Hash algorithms can be used to make blockchain integration safe, and many of them can solve problems with data integrity and security that arise when using blockchain technology. But they can also have problems with time, lack of resources, and using too much memory. In this study, an algorithm to create a hash based on a chaos key using logistic maps and lightweight cryptography is proposed (hamming bird 2). Hash outputs are tested in terms of time and hardness to guess using cryptanalysis tools. The proposed algorithm has been tested using cryptanalysis site based brute force attacks. The result show that the modified hash cannot identified by site using all cryptanalysis setting that prove the strong of hash generated against attacks.

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1. Introduction

Blockchain is one of the newest technologies in the area of security, traceability, and openness. It can be used to manage any transaction involving a digital asset, in addition to physical assets and agreements. In blockchain

* Corresponding author. Tel.: +0-000-000-0000 ; fax: +0-000-000-0000.
E-mail address: author@institute.xxx

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technology, users can't change or delete any transactions that have already happened. So, all transactions that have happened in the past are clear and can't be changed[1][2]. Blockchains, in contrast to the web or internet by itself, are decentralized rather than centralized, open rather than hidden, inclusive rather than exclusive, immutable rather than mutable, and secure. [3][4].

The capabilities offered by blockchain for the production and exchange of value in society are unprecedented. In its most basic form, a blockchain is a chain of blocks, and each block is made up of three components: data, a hash of the data, and the hash of the block that came before it. Each individual block in the chain stores its own unique cryptographic hash as well as the hash of the most recent block that was necessary for the chain to remain connected. Blocks are the fundamental building blocks of a blockchain.[5][6]. A block is a group of data or information in the blockchain. The data is inserted into the block on the blockchain by linking it with other blocks in time order, so forming a chain of blocks. Consequently, it creates a shared, historical database of transactions among several nodes, that is, computers or servers, in a network[4]. To generate the next block in the chain, miners choose a unique number, called a Nonce, which is appended to a hashed or encrypted block and may only be used once in the process. In other words, it's a form of "Proof of Work."[7][8]. When a transaction takes place in the blockchain, a hash is created to uniquely identify it. A hash is calculated using a combination of new data, the previous hash, and the current time. Each block in a blockchain provides a permanent record of the most recent transaction that occurred there (Figure 1) [9]Before adding a block to a chain, its validity must be verified via a consensus mechanism in which the majority of nodes (clients or servers) and the nodes with the highest stack in the distributed network's chain validate the block. After the block has been validated, a unique identifier, or hash, is generated. This eliminates the requirement for a third party to validate or conduct transactions. [10]. Blocks can be identified by their block numbers, block heights, and the hashes of their block headers. A piece of software known as the hash function is used to do the verification of the data contained within the blocks. This function ensures that the data cannot be altered in any way and prevents it from being seen by the participants in the Blockchain. Each block is responsible for its own hash function. In a blockchain, data is preserved for the foreseeable future and cannot be altered in any way. Any variation at all in it will cause the generation of a new block in the chain. The blockchain functions much like a digital notary, complete with timestamps to prevent any information from being altered. Because the vast majority of hash functions are one-way functions, it is extremely difficult to locate an input which hashes to a certain hash value[11]. Those hash functions that do not require a secret key are referred to as "un-keyed hash functions," and those that do require a secret key are referred to as "keyed hash functions" (known as "keyed hash functions") which require both a message and a secret key in order to perform properly[12][13]. The hash function Keccak256, that is utlized in the Ethereum blockchain, represent a variant of SHA3_256 with various constants changed in the code. The SHAKE_128 hash functions and SHAKE_256 128 hash functions are variants of the SHA3_256 and SHA3_512 algorithms, inwhich the output message length can vary. SHA-3 is the most recent secure hash algorithm technique that the NIST has agreed on[14]. SHA-3 is a good way to make sure that a transaction between a sender and a receiver is authentic. The sender's message looks like this: (hash code, data). The "hash code" is the unique signature for the communication[15].

In this work, we have been proposed a new method to modify SHA3 hashing algorithm by combined two cryptography techniques, one for securing communication that utilized modified SHA-3 hash algorithm and the second is to secure data transferred by utilized light weight encryption.

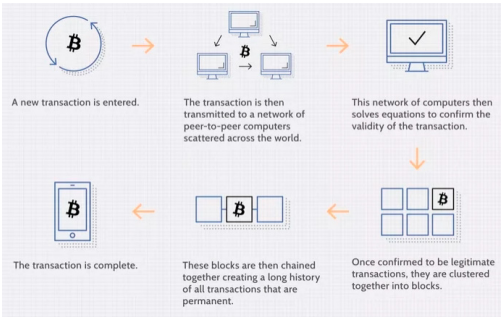

Fig. 1. The Blockchain Technology [32]

2. Secure Hashing Blockchain

SHA-1, SHA-2, and SHA-256 are the algorithms that provide the greatest level of safety and dependability in block chain technology [16], [17]. A superior hash function is used. It is possible to generate a unique output based on a number of different inputs, and this result serves as a transaction-specific key. The SHA was created by the US Federal Information Processing Standard and the (NSA)[18]. Due to the reliability of blockchain functions, an appropriate and secure hash code can be generated. All proposed blockchain transactions are hashed before being recorded. All of the Blocks are linked to one another, and the hash indicators store the hash values from the previous block[19] [33]. Both SHA1 and SHA2 are variants of the same secure hash algorithm, but they differ in syntax (how to build the final hash from the original data) and signature bit length, which means that any modifications to a blockchain transaction will affect all of the blocks involved[20]. It is proved that SHA2, the algorithm that followed SHA1, provides a serious advancement[21]. In terms of the number of bits, SHA1 has 160 bits, while SHA 2 can have a number of different lengths, the most popular of which is 256 bits. SHA 224, SHA 384, and SHA 512 are all variations of the SHA2 hashing algorithm[22]. Between the years 2011 and 2015, the most common algorithm was called Sha1. On the other hand, a growing corpus of research has shown that SHA1 has flaws in relation to collisions (i.e., a couple pieces of data originated the same hash)[18]. The hashing algorithm must satisfy the following conditions: [23]
1- eterministic: The hash does not change for the same message—that is, considering a certain input M, the function always calculates same output $h(M)$.
2- Quick computation: Calculation speed.
3- Pre-image resistance: This refers to the impossibility of deducing the input through the hash. That is, considering hash value H , it's infeasible M computationally to find input such that $.h(M) = H$.
4- A small change in input changes the hash: A small change in entered data leads to a change of the hash.
5- Collision resistant: No two messages have the same hash M and $\acute{M}$ such that $h(M) = h(\acute{M})$ and $M \neq \acute{M}$. which has been referred to as strong collision resistance as well.
6- Hashtags should be non-modifiable, not just encrypted.
7- No correlation: input bits M should not be connected to output bits $h(M)$.
8- Random behaviour: Random behaviour has to be included in the hash function, making it useless to attempt to predict which output is associated with which input.
Fig. 2(a) illustrates how blockchain technology may be conceptualized as the combination of a block that is a part of its structure and consists of data with the hash of the block that came before it (except for the original one, that doesn't have a previous hash). Figure 2b depicts a blockchain network that consists of numerous blocks, each of which is hash-linked to the block that came before it.

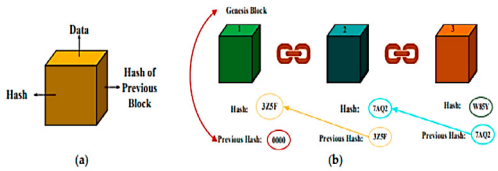

Fig. 2. (a) Hash block component. (b) Block-chain network [34].

SHA1 launched the National Institute of Standards and Technology (NIST) in 1995, and it was adopted for many years[24]. Numerous researchers have demonstrated that it is feasible to receive contradictory messages under specific situations. Consequently, SHA1's level of safety drops. The Sha2 hash was announced in 2001, however it did not live up to expectations[25]. Therefore, hash algorithm development is ongoing. The Institute announced an open debate on a newer hash function, SHA3, through October 2008[26]. Thus, researchers always look for hash algorithms that ensure their first notion, that they are not reversible, and their second concept, that it is difficult to discover two inputs that produce the same hash value.

3. Methodology

SHA3 Algorithm SHA3 is the Extendable-Output Functions and Permutation-Based Hash which is specified in FIPS202. These hashes are available in four types, provides SHA-3-224, SHA-3-256, SHA-3-384 and SHA-3-512[27].In python language, "hashlib" library provides all hashes from FIPS 202. In this work we used SHA3-256 as an encryption function for password.However, even though SHA-3 is very strong, hashed passwords are not different to themselves because to the deterministic nature of the hash function: while given the same input, the same result will always be returned. Hence, the attacker can more accurately estimate the password that corresponds to that hash. Once the password is known, it can be used to access any accounts that using that hash.To overcome this weakness, salt has been used. Salt represents a value produced by a cryptographically secure function which can be integrated with the input of hash functions to generate special hashes for each input, no matter of the input not is unique. A salt can make a hash function appear non-deterministic, and that is good as it cannot expose duplicate passwords via our hashing[28]. This hash algorithm scheme employs the following steps to secure the WSN, password:

First Modified SHA3 (FMSHA3) using Chaotic Key Algorithm

For password encryption algorithm, we have utilized chaotic map which is the one of the most effective methods of encryption having an extreme sensitivity of its starting condition. There are several types of chaotic map, one of the common types is the logistic map, which is a 2-D map full of chaos. There have been different ideas for logistic maps, all of which are possible. But to choose one, it must have three qualities, such as Mixing Property, Robust Chaos, and Large Parameter. We're going to use a traditional logistic map[29] to look at all of the properties. The answer is:

$$x_{i+1} = rx_n(1 - x_n) \tag{1}$$

Where, X_i lies between zero and one, and the interval for r is $[0; 4]$

$$g(x) = 4x(256 - x)/256 \tag{2}$$

From initial condition. Then, transform sequence into another one is done by:

$$h_1(x) = \left\lfloor x \cdot 10^6 \right\rfloor (256 - X) \bmod 256 \tag{3}$$

This is a key for encryption as well decryption with size 256 bits. The key can generate a sequence of integer that we then integrate it with SHA-3 hash value. The combination is done be converting SHA3 value to hexadecimal Base16 and then summation with chaos key to produce new value then convert to hex. The combination password algorithm scheme employs the following steps to secure the WSN, password:

Algorithm (1) Modified SHA3
Input: Message=Hashed (By SHA3+Salt)
Output: Modified Hash
Begin
Step 1: Read Message M as bytes.
Step 2: Rotate the message 5 time to right.
Step 3: Apply XORing between Shifted Ms and K and get MS.
Step 4:using keccak function to compute the final hash:
NH=keccak (MS, 1600, 1600-2*n, n, K).
Step 5: Save hash NH.
End.

After modified hash is generated it send to WSN node which start same process with saved password then it comperes

between the received and saved if the result is true then authentication success and node will combine with WSN and then can transfer file to others.

HammingBird2 Lightweight Algorithm

Hummingbird-2 is a basic form of encryption made for devices with limited resources, like RFID tags, wireless sensors, smart meters, and industrial controllers. Hummingbird-2 can be made to work with very little hardware or software, so it can be used to protect low-cost devices that are everywhere[30].

The proposed modified Hummingbird-2 (MHB2) cipher has a 128-bit secret key K and a 128-bit internal state R which is initialized using a 64-bit Initialization Vector IV. These variables are accessed as vectors of 16-bit words[31]:

$$K = (K_1, K_2, K_3, K_4, K_5, K_6, K_7, K_8) \tag{4}$$

$$R = (R_1, R_2, R_3, R_4, R_5, R_6, R_7, R_8) \tag{5}$$

$$IV = (IV_1, IV_2, IV_3, IV_4) \tag{6}$$

Modified Hummingbird-2 is entirely built from operations on 16-bit words: the exclusive-or operation on words ($\bigoplus$), addition modulo 65536 ($\boxplus$) and a nonlinear mixing function $f(x)$.

WD16(x, a, b, c, d) and Nonlinear functions f(x) =

The nonlinear mixing function f consists of four-bit S-Box permutation lookups on each nibble of the word, followed by a linear mix. Table 1 shows the modified Hummingbird- 2 SBoxes .

Table 1. S-Boxes used in Hummingbird-2.

X	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
S1(X)	7	12	14	9	2	1	5	15	11	6	13	0	4	8	10	3
S2(X)	4	10	1	6	8	15	7	12	3	0	14	13	5	9	11	2
S3(X)	2	15	12	1	5	6	10	13	14	8	3	4	0	11	9	7
S4(X)	15	4	5	8	9	7	2	1	10	3	0	14	6	12	13	11

Let S(x) denote the computation of four S−Boxes and L(x) the linear transformation which is expressed using the left circular shift (rotation) operator ($\ll$). We may write the f component as

$$S(x) = S_1(x_0) | S_2(x_1) | S_3(x_2) | S_4(x_3) \tag{7}$$

$$L(x) = x \oplus (x \ll 6) \oplus (x \ll 10) \tag{8}$$

$$f(x) = L(S(x)) \tag{9}$$

We further define a 16-bit keyed permutation WD16 using f as

$$WD16(x, a, b, c, d) = f(f(f(f(x \oplus a) \oplus b) \oplus c) \oplus d) \tag{10}$$

The inverse of $f(x)$ and WD16 can be derived in straightforward fashion.

Initialization

The internal state of Hummingbird-2 is initialized with a four-round procedure using the 64-bit nonce IV. We first set.

$$R^{(0)} = (IV_1, IV_2, IV_3, IV_4, IV_1, IV_2, IV_3, IV_4) \text{ and}$$

$$P^{(0)} = (P1, P2, P3, P4) \tag{11}$$

Where P1 is first 256-bit block of generated chaos key, P2 is second 256-bit block of generated chaos key, P3 is third 256-bit block of generated chaos key, P4 is fourth 256-bit block of generated chaos key. and then iterate for $i = 0, 1, 2, 3$ the following:

$$t_1 = WD16\left(R_1^{(i)} \boxplus \langle i \rangle, Pk1, Pk2, K_3, K_4\right) \tag{12}$$
$$t_2 = WD16\left(R_2^{(i)} \boxplus t_1, K_5, Pk2, Pk3, K_8\right) \tag{13}$$
$$t_3 = WD16\left(R_3^{(i)} \boxplus t_2, K_1, K_2, Pk2, Pk4\right) \tag{14}$$
$$t_4 = WD16\left(R_4^{(i)} \boxplus t_3, Pk1, K_6, K_7, Pk4\right) \tag{15}$$
$$R_1^{(i+1)} = \left(R_1^{(i)} \boxplus t_4\right) \ll 3 \tag{16}$$
$$R_2^{(i+1)} = \left(R_2^{(i)} \boxplus t_1\right) \gg 1 \tag{17}$$
$$R_3^{(i+1)} = \left(R_3^{(i)} \boxplus t_2\right) \ll 8 \tag{18}$$
$$R_4^{(i+1)} = \left(R_4^{(i)} \boxplus t_3\right) \ll 1 \tag{19}$$
$$R_5^{(i+1)} = \left(R_5^{(i)} \oplus R_1^{(i+1)}\right) \tag{20}$$
$$R_6^{(i+1)} = \left(R_6^{(i)} \oplus R_2^{(i+1)}\right) \tag{21}$$
$$R_7^{(i+1)} = \left(R_7^{(i)} \oplus R_3^{(i+1)}\right) \tag{22}$$
$$R_8^{(i+1)} = \left(R_8^{(i)} \oplus R_4^{(i+1)}\right) \tag{23}$$

The initial state for encrypting the first plaintext word is $R^{(4)}$. Hence, that the two's complement numerical value of $(i = 0 \ldots 3)$ is used in computation of t_1.

Encryption

Encryption of a single word of plaintext P_i to ciphertext word C_i requires four invocations of $WD16$.

$$t_1 = WD16\left(R_1^{(i)} \boxplus P_i, Pk_1, Pk_2, K_3, K_4\right) \tag{24}$$
$$t_2 = WD16\left(R_2^{(i)} \boxplus t_1, K_5 \oplus R_5^{(i)}, Pk2 \oplus R_6^{(i)}, Pk3 \oplus R_7^{(i)}, K_8 \oplus R_8^{(i)}\right) \tag{25}$$
$$t_3 = WD16\left(R_3^{(i)} \boxplus t_2, K_1 \oplus R_5^{(i)}, K_2 \oplus R_6^{(i)}, Pk2 \oplus R_7^{(i)}, Pk4 \oplus R_8^{(i)}\right) \tag{26}$$
$$C_i = WD16\left(R_4^{(i)} \boxplus t_3, Pk1, K_6, K_7, Pk4\right) \boxplus R_1^{(i)} \tag{27}$$

After each encrypted / decrypted word, we perform the following state update:

$$R_1^{(i+1)} = \left(R_1^{(i)} \boxplus t_3\right) \tag{28}$$
$$R_2^{(i+1)} = \left(R_2^{(i)} \boxplus t_1\right) \tag{29}$$
$$R_3^{(i+1)} = \left(R_3^{(i)} \boxplus t_2\right) \tag{30}$$
$$R_4^{(i+1)} = \left(R_4^{(i)} \boxplus R_1^{(i)} \boxplus t_3 \boxplus t_1\right) \tag{31}$$
$$R_5^{(i+1)} = R_5^{(i)} \oplus \left(R_1^{(i)} \boxplus t_3\right) \tag{32}$$
$$R_6^{(i+1)} = R_6^{(i)} \oplus \left(R_2^{(i)} \boxplus t_1\right) \tag{33}$$
$$R_7^{(i+1)} = R_7^{(i)} \oplus \left(R_3^{(i)} \boxplus t_2\right) \tag{34}$$

$$R_8^{(i+1)} = R_8^{(i)} \oplus \left(R_4^{(i)} \boxplus R_1^{(i)} \boxplus t_3 \boxplus t_1\right) \tag{35}$$

This can be summed up using the formula $C = E(P)$. Each time the function E is called, the value of the state variable R advances by one iteration. Take note that the update function could be streamlined as a result of the repeated use of certain phrases.

Decryption

Decryption of a single word $C_i \in F, i = 1, 2, ...,$ followed by the same initialization, is:

$$u_3 = WD16^{-1}\left(C_i \boxminus R_1^{(i)}, Pk4, K_7, K_6, Pk1\right) \tag{36}$$

$$u_2 = WD16^{-1}\left(u_3 \boxminus R_4^{(i)}, Pk2 + R_8^{(i)}, Pk2 + R_7^{(i)}, K_2 + R_6^{(i)}, K_1 + R_5^{(i)}\right) \tag{37}$$

$$u_1 = WD16^{-1}\left(u_2 \boxminus R_3^{(i)}, K_8 + R_8^{(i)}, Pk4 + R_7^{(i)}, Pk2 + R_6^{(i)}, K_5 + R_5^{(i)}\right) \tag{38}$$

$$P_i = WD16^{-1}\left(u_1 \boxminus R_2^{(i)}, K_4 + K_3+, Pk2+, Pk1\right) \tag{39}$$

After this, the internal states are updated as in the encryption, i.e., using Eq. (3.28 -3.34), were:

$$t_3 = u_3 \boxminus R_4^{(i)} \tag{40}$$

$$t_2 = u_2 \boxminus R_3^{(i)} \tag{41}$$

$$t_1 = u_1 \boxminus R_2^{(i)} \tag{42}$$

Modified SHA3 by Hybrid of FMSHA3 and MHB2 (SMSHA3)

The encrypted data resulted from the MHB2 algorithm with divided into blocks. These blocks passed to the FMSHA3 hashing algorithm. We proposed a modification to the SHA3 algorithm (by using chaos key as explain in 3.3) and MHB2 algorithm (with 2 rounds only) to increase the randomness and FMSHA3 security. Algorithm (2) shows the modifications to the SMSHA3. First modification is to shifted the message blocks by 3 cycles to left and XORed with chaos key K (first digits of the key) in order to increase the randomness of the message blocks. Second modification is to use the chaos key K and MHB2 algorithm to generate the KecHash that used in the Keccak function. Keccak function (explained in appendix A) will compute the SMSHA3-256 hash value.

Algorithm (2) Modified SHA3-256 Hashing Algorithm (SMSHA3).
Input: message Ms, K5, K6, n=256.
Output: hash value.
Begin
Step 1: Read Message M as bytes.
Step 2: shift the message 3 time to left.
Step 3: Apply XORing between Shifted Ms and K and get MS.
Step 4: Apply the MHB2 algorithm (use 2 rounds) with input
data(MS) and chaos key(k) and get HF.
Step 5: compute keck-hash using the MHB2 algorithm (use 2 rounds) with input data (K) and shifted by 8 to right.
Step 6: using keccak function to compute the final hash:
FH=keccak (HF, 1600, 1600-2*n, n, keck-hash).
Step 7: Save hash FH.

4. Results and Discussion

In this section we investigate the process of modified hashing of blockchain.Generates Chaotic Key Series and Combined with Hashed Password.In this part, chaos keys set are generated based on equation described in methodology using 4 control parameters. Table 1 show the chaos key set results with different control parameters.

Table 2. Chaos key set results with different control parameters

Key	Control Parameters			Password Hash Value
	x	r	s	
K1	0.25	0.685	92	[0, 51, 101, 0, 126, 159, 249, 0, 115, 212, 129, 23, 236, 111, 112, 2, 217, 215, 14, 8, 224, 246, 187, 211, 117, 83, 196, 52, 64, 21, 124, 202, 33, 151, 17, 171, 18, 23, 181, 164, 72, 251, 252, 3, 69, 9, 162, 202, 156, 139, 252, 96, 106, 212, 68, 129, 88, 193, 180, 169, 177, 165, 125, 37, 70, 204, 221, 40, 102, 7, 250, 133, 38, 135, 106, 167, 34, 198, 136, 93, 63, 43, 29, 20, 14, 9, 6, 4, 3, 2, 1, 0]
K2	0.45	0.75	92	[0, 116, 109, 107, 185, 125, 62, 252, 57, 27, 238, 13, 104, 133, 100, 210, 170, 11, 41, 219, 4, 146, 191, 85, 38, 201, 168, 55, 119, 244, 8, 133, 113, 254, 219, 132, 10, 139, 14, 44, 27, 70, 160, 184, 78, 101, 16, 146, 225, 25, 102, 180, 249, 242, 241, 242, 116, 86, 128, 224, 168, 61, 46, 162, 250, 187, 140, 105, 79, 251, 124, 29, 214, 224, 104, 142, 42, 224, 40, 30, 214, 32, 152, 242, 117, 152, 178, 197, 84, 63, 175, 3]
K3	0.45	0.685	50	[0, 11, 53, 42, 111, 38, 55, 102, 252, 155, 54, 219, 149, 220, 235, 177, 140, 46, 209, 249, 46, 241, 218, 243, 6, 166, 247, 106, 33, 24, 220, 185, 129, 182, 214, 210, 180, 28, 84, 117, 228, 213, 205, 240, 125, 37, 181, 234, 246, 54]
K4	0.85	0.85	99	0, 140, 31, 43, 22, 42, 140, 241, 163, 15, 120, 244, 32, 242, 223, 234, 191, 230, 228, 203, 117, 192, 89, 200, 18, 72, 78, 59, 224, 191, 6, 182, 224, 201, 251, 200, 254, 233, 168, 72, 19, 1, 180, 190, 104, 132, 162, 182, 220, 182, 198, 92, 70, 166, 157, 56, 65, 253, 87, 238, 215, 221, 7, 171, 111, 94, 15, 75, 147, 121, 198, 110, 32, 45, 17, 90, 152, 40, 213, 85, 152, 119, 187, 214, 31, 199, 162, 114, 221, 86, 77, 200, 76, 151, 49, 78, 219, 223, 176]

The keys change depending on the parameter. Each key is combined with a related hash (i.e., combined k1 with hash 1, k2 with hash 2 etc.). Table 3 shows the modified hash values obtained by combining chaotic key set results and hashed results.

Table 3. Hash value for WS password after combined with chaotic key set

WS ID	Password Hash Value
R1	13236e3c49da2626b18b33c9c042c10a970ede33ce3cdad37b7591 037ad60edea673632e6da521eea41b5f0ae0082e55ba256a6fb413b c4fd7df5feb24686bfec515f2df9fa994794244e658293536552f6c3 1a7a5e854dadaf43e55d731e85108dcbe61f92fbe6961d3
R2	13236e3c49da2626b18b33c9c042c10a970ede33ce3cdad37b7591 037ad60edea673632e6da521eea41b5f0ae0082e55ba256a6fb413b c4fd7df5feb24686bfec515f2df9fa99479424
R3	13236e3c49da2626b18b33c9c042c10a970ede33ce3cdad37b7591 037ad60edea673632e6da521eea41b5f0ae0082e55ba256a6fb413b c4fd7df5feb24686bfec515f2df9fa99479424584de3ce410f5126ca acb7ab28c23f1825b69f90e717ef8dd6d0b604ac39fba8e
R4	13236e3c49da2626b18b33c9c042c10a970ede33ce3cdad37b7591 037ad60edea673632e6da521eea41b5f0ae0082e55ba256a6fb413b c4fd7df5feb24686bfec515f2df9fa9947942451e8108f8e2cbf698a 7d3dbde00e5de5bf1c517b61cf06eddb9640d75d3fd895b

From table 3, the new hashed in R1 is 212 bytes in format of hexadecimal string, so it very hard or even impossible to attacker that using cryptanalysis tools to determine the hash algorithm that used and guess the original blockchain hash value. Compared with standard hash used in blockchain, the model is more secure and it very hard to cracked due to several point of strength which are:

1- The proposed model has utilized the SHA-3 (Keccak) hashing algorithm which is the newest strong hash algorithm that is the most accurate and high secure hashing algorithms that expected it could not be cracked in the next 20 - 30 years.
2- Integrate Chaotic Key set with original hash: this make hash harder to guess or determined the hash function as the hash value changed where the hash value output is highly depends on three control parameters.

In summary, the attacker needs first to identify how combination algorithm work, need to get salt value, need to have exact chaotic control parameters to retrieve original SHA3 value which is also not broken till now. Next, we have tested the modified hash under brute force attack and compared it with other state of art approaches where we used most powerful cracking site ”onlinehashcrack.com ” This site has a wide range of hashing algorithms and option to control brute force attack. This site is used for scientific tests as well as used by some attackers. For test we test most common state of art hash algorithm such as MD5, SHA1, SHA2, SHA3.

Table 4. Hash value for different hash algorithms using same

Hash Algorithm	Password Hash Value
MD5	202cb962ac59075b964b07152d234b70
SHA1	40bd001563085fc35165329ea1ff5c5ecbdbbeef
SHA-256	a665a45920422f9d417e4867efdc4fb8a04a1f3fff1fa07e9 98e86f7f7a27ae3
SHA3-256	a03ab19b866fc585b5cb1812a2f63ca861e7e7643ee5d43 fd7106b623725fd67
Modified SHA3-256	13236e3c49da2626b18b33c9c042c10a970ede33ce3cdad 37b7591037ad60edea673632e6da521eea41b5f0ae0082e 55ba256a6fb413bc4fd7df5feb24686bfec515f2df9fa9947 94244e658293536552f6c31a7a5e854dadaf43e55d731e8 5108dcbe61f92fbe6961d3

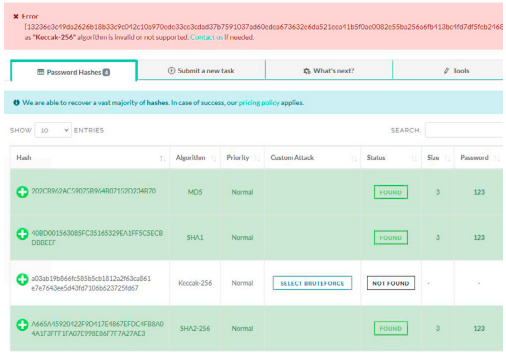

Fig. 3. The cracked results for different hash value from different hash algorithms.

From cracking process, is successfully found value from MD5, SHA1, SHA256 hash in very fast less than a second. However, it determines the hash type SHA3 (Keccak) but it failed to found the password from this hash after 1 week. The site failed to determine the hash type of modified hash even select all available algorithms and show error message as shown in figure 3.

5. Conclusions

This work mainly emphasizes developing a modified hashing algorithm for SHA-3 256 algorithm. To achieve this goal, three process has been combined to generate a high-level hashed value. This technique includes, hashing using traditional SHA-3 256 then adding chaotic 2D key by generated general sequence keys that will added to hashed key to produce new hash which is different in length and value from original hash. The new method is assumed to be hard to cracked by any crack program this belong to the following point of strength.

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